MEDIUM ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

01

10 questions · ~15 min

Q1 ADVANCED MATH

If $4\sqrt{2x} = 16$, what is the value of $6x$?

A. 24

B. 48

C. 72

D. 96

Q2

ADVANCED MATH

$$5x^2 - 37x - 24 = 0$$

What is the positive solution to the given equation?

A. $\frac{3}{5}$

B. 3

C. 8

D. 37

$$x^2 = 6x + y$$
$$y = -6x + 36$$

A solution to the given system of equations is (x, y). Which of the following is a possible value of xy ?

A. 0

B. 6

C. 12

D. 36

Q4

ADVANCED MATH

If $\frac{42}{x} = 7x$, what is the value of $7x^2$?

A. 6

B. 7

C. 42

D. 294

Q5 ADVANCED MATH

$$14j + 5k = m$$

The given equation relates the numbers j, k, and m. Which equation correctly expresses k in terms of j and m?

A. $k = \frac{m - 14j}{5}$

B. $k = \frac{1}{5}m - 14j$

C. $k = \frac{14j - m}{5}$

D. $k = 5m - 14j$

Q6 ADVANCED MATH

$$-4x^2 - 7x = -36$$

What is the positive solution to the given equation?

A. $\frac{7}{4}$

B. $\frac{9}{4}$

C. 4

D. 7

Q7 ADVANCED MATH

$$T = 0.01(P - 40{,}000)$$

In a city, the property tax T, in dollars, is calculated using the formula above, where P is the value of the property, in dollars. Which of the following expresses the value of the property in terms of the property tax?

A. $P = 100T - 400$

B. $P = 100T + 400$

C. $P = 100T - 40{,}000$

D. $P = 100T + 40{,}000$

Q8

ADVANCED MATH

$$5x + 42x - 5 = 0$$

Which of the following is a solution to the given equation?

A. $-\frac{5}{2}$

B. $-\frac{5}{4}$

C. $-\frac{4}{5}$

D. $-\frac{2}{5}$

Q9

ADVANCED MATH

$$p = 20 + \frac{16}{n}$$

The given equation relates the numbers p and n, where n is not equal to 0 and $p > 20$. Which equation correctly expresses n in terms of p?

A. $n = \frac{p - 20}{16}$

B. $n = \frac{p}{16} + 20$

C. $n = \frac{p}{16} - 20$

D. $n = \frac{16}{p - 20}$

Q10 ADVANCED MATH

$$y = x^2 - 1$$
$$y = 3$$

When the equations above are graphed in the xy-plane, what are the coordinates (x, y) of the points of intersection of the two graphs?

A. $(2,3)$ and $(-2,3)$

B. $(2,4)$ and $(-2,4)$

C. $(3,8)$ and $(-3,8)$

D. $(\sqrt{2},3)$ and $(-\sqrt{2},3)$